Do Reward Models Encode What Their Labels Claim?

An Interpretability Audit of ArmoRM

Sathammai

github.com/Sat1509/reward-basis-audit

***Abstract*—ArmoRM decomposes reward into 19 explicitly labeled heads — helpfulness, safety, honesty, and 16 others. On naturalistic preference data (hh-rlhf), 1 of 19 heads reliably predicts human judgment above chance. The valid one is readability, not helpfulness or safety. On two additional datasets, 17–18 of 19 heads work well. Same model; completely different picture. The core finding: whether a labeled reward decomposition appears interpretable depends entirely on whether the evaluation data isolates the signal being measured. This distinction matters significantly for how we audit reward models.**

Epistemic note: empirical results are real. Interpretive claims about safety are more speculative.

I. Background

Modern RLHF trains a reward model on human preferences, then fine-tunes the LLM to maximize that reward. The safety question: what is the reward model actually measuring?

Most reward models are opaque scalars. ArmoRM (`RLHFlow/ArmoRM-Llama3-8B-v0.1`) decomposes reward into 19 explicitly labeled components — helpfulness, safety, honesty, coherence, correctness, and 14 more — combined by a learned gating network into a final scalar.

The premise is *transparency*. If the safety head can be audited — if it verifiably responds to safety rather than surface noise — you have a genuine inspection tool. If not, the labels are a UI on a black box: interpretability theater.

Question: do the labeled reward heads correspond to what the model actually learned?

II. What ArmoRM Is

On top of Llama 3 8B: a $\texttt{Linear}(4096 \to 19)$ regression layer producing 19 raw head scores, and a small MLP gating network outputting 19 context-dependent mixing weights. Final reward = dot product of weights and scores.

ArmoRM's paper validates the *final scalar*. I audit the *individual heads* — a strictly harder test, since the gating network can rescue a combined score by routing around noisy individual heads.

III. Experiments

Four analyses over 800 chosen/rejected pairs from `Anthropic/hh-rlhf`, then extended to two additional datasets.

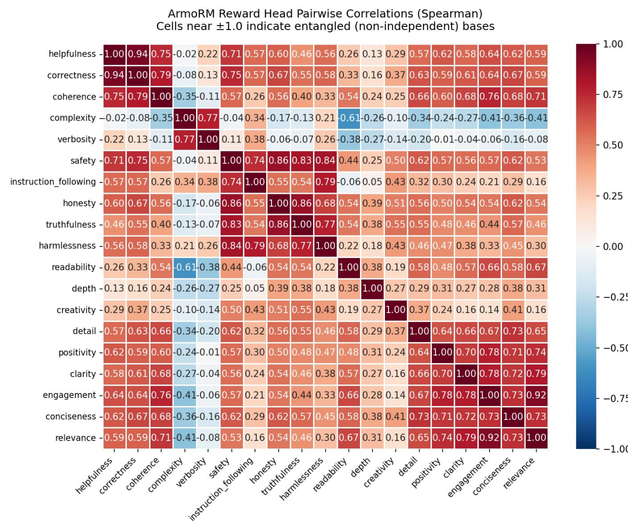

Fig. 1: Spearman rank correlations between all 19 reward heads (1600 responses). Safety, honesty, harmlessness, and truthfulness form one tight cluster ($r = 0.83$–0.86).

A. Analysis 1: Are the Heads Independent?

Spearman rank correlation across all 171 head pairs, 1600 responses pooled. 25 pairs exceeded $|r| > 0.7$; mean absolute off-diagonal correlation: 0.454. The 19 heads collapse into $\sim$5 groups.

TABLE I: Selected high-correlation pairs ($|r| > 0.7$).

Head pair	**Spearman r**
helpfulness ↔ correctness	0.943
engagement ↔ relevance	0.915
honesty ↔ truthfulness	0.863
safety ↔ honesty	0.863
safety ↔ harmlessness	0.836
safety ↔ truthfulness	0.826
complexity ↔ verbosity	0.773

Safety, honesty, truthfulness, and harmlessness form one cluster. Upweighting the safety head in an RLHF objective also upweights honesty and harmlessness by proxy — the decomposition implies independence but does not deliver it.

B. Analysis 2: Do the Heads Encode Anything Real?

Pooled last-token hidden states (4096-dim) reduced to 50 dims via PCA (79.8% variance retained), logistic regression probe per head (5-fold CV, chance = 0.50).

TABLE II: Linear probe accuracy, selected heads.

Head	Probe accuracy
engagement	0.923 ± 0.011
readability	0.905 ± 0.011
safety	0.877 ± 0.012
helpfulness	0.869 ± 0.020
depth	0.738 ± 0.013

All 19 heads probe strongly — every dimension is linearly encoded. The model learned something real for each head. This sets up the central tension: if representations are real, why do heads fail the validity test?

C. Analysis 3: Where Is the Entanglement?

Regression of behavioral co-firing (score correlation) onto geometric overlap (cosine similarity in $W \in \mathbb{R}^{19\times 4096}$):

$$\text{Spearman}(s_i, s_j) = 1.213 \cdot \cos(w_i, w_j) + 0.273, \quad R^2 = 0.28$$

Weight vectors are nearly orthogonal (gap: 0.038). Only 28% of co-firing explained by weight geometry — the regression layer is not the source. The entanglement lives in the backbone H: both w_{safety} and $w_{\text{helpfulness}}$ act on the same H, from a backbone where safety and helpfulness almost always co-vary. The fix is not in W; it is in H.

D. Analysis 4: Do the Heads Track Human Judgment?

Fraction of pairs where each head scores the chosen response higher (0.50 = chance). Wilcoxon signed-rank test on the gap distribution. **On `hh-rlhf`: 1 of 19 heads valid.**

TABLE III: Preference accuracy on `hh-rlhf`. Validity threshold: 0.65.

Head	Acc.	Sig.
readability	0.711	$*$
depth	0.613	$*$
truthfulness	0.599	$*$
honesty	0.568	$*$
safety	0.520	$*$
harmlessness	0.525	$*$
helpfulness	0.429	ns
complexity	0.380	ns
verbosity	0.390	ns

The helpfulness head predicts the wrong response 57% of the time on data where annotators explicitly chose the more helpful response. The harmlessness head scores 0.525 on harmlessness-subset pairs specifically — near coin flip even where the annotation criterion matched the label.

Is this a model failure or a dataset problem? `hh-rlhf` is naturalistic — even in the harmlessness subset, the chosen response is likely also more helpful, more coherent, better written. A head can only demonstrate its labeled concept when the test actually isolates that concept.

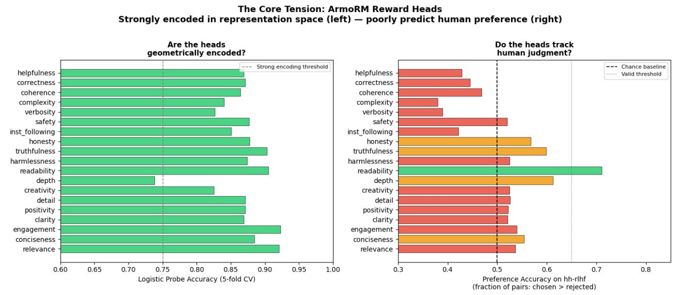

Fig. 2: Every head encodes something real geometrically (left, probe accuracy) yet most fail to track human judgment (right, preference accuracy on `hh-rlhf`). The gap between panels is the central finding.

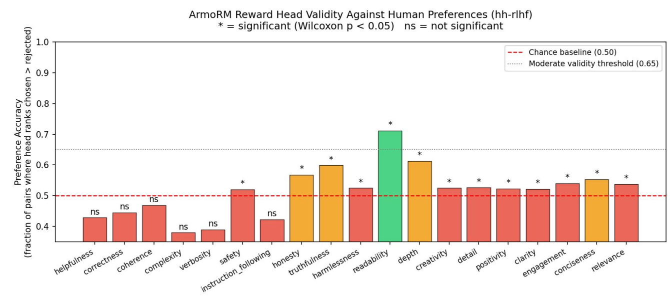

Fig. 3: Per-head preference accuracy on `hh-rlhf` ($N = 800$). Only readability clears 0.65. Helpfulness and safety sit at 0.429 and 0.520.

IV. Extending to Reward Bench

`allenai/reward-bench`: 2985 pairs across 23 capability-specific subsets. **17 of 19 heads valid. Mean accuracy 0.79.** Only complexity (0.504) and verbosity (0.539) fail.

TABLE IV: Safety head accuracy by Reward Bench subset.

Subset	Accuracy
refusals-dangerous	0.870
refusals-offensive	0.950
xstest-should-refuse	0.864
donotanswer	0.603
llmbar-adver-GPTInst	0.500

Safety head reaches 0.87–0.95 where safety is the explicit criterion, drops to chance on adversarial instruction-following pairs. However, Reward Bench was used in ArmoRM's original paper — distribution overlap is a concern. An independent dataset is needed.

V. The Decisive Experiment: HelpSteer2

`nvidia/HelpSteer2`: per-response human ratings on helpfulness, correctness, coherence, complexity, verbosity (0–4 Likert), collected independently of ArmoRM. These five names map directly onto ArmoRM heads. Multiple responses per prompt allow two pair types:

- **TYPE A (overall):** one response is generally better across all dims — mirrors `hh-rlhf`.

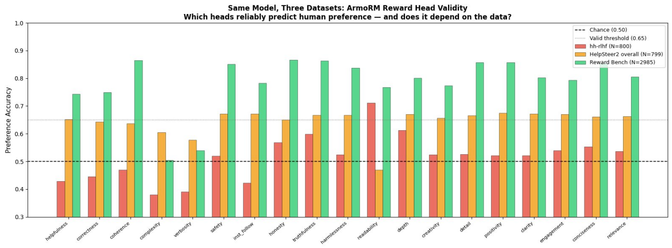

Fig. 4: All 19 heads across all three datasets. Same model, same weights — completely different validity picture. Dataset choice, not model quality, drives the outcome.

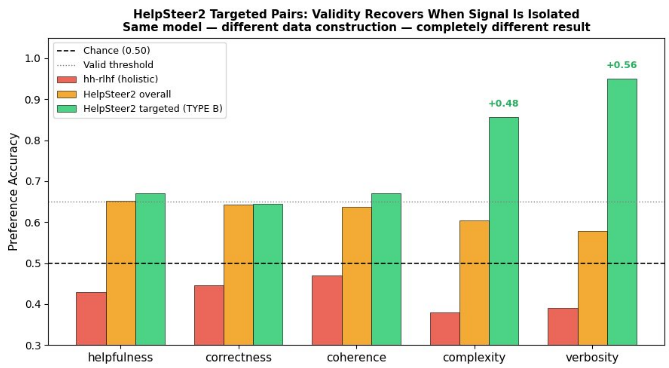

Fig. 5: The 5 HelpSteer2-matched heads across evaluation conditions. Complexity (+0.48) and verbosity (+0.56) show the sharpest recovery from `hh-rlhf` to TYPE B pairs.

- **TYPE B (targeted):** target dimension gaps ≥ 2; all others gap ≤ 1. Exactly one dim varies; rest controlled.

TYPE A (799 pairs): 18 of 19 valid. Invalid: readability (0.469) — the only valid head on `hh-rlhf`, now invalid. This directly reveals what `hh-rlhf` was actually measuring.

TABLE V: Head accuracy across conditions.

Dim.	hh-rlhf	HS2 A	HS2 B
helpfulness	0.429	0.652	0.670
correctness	0.445	0.643	0.645
coherence	0.469	0.637	0.671
complexity	0.380	0.605	**0.857**
verbosity	0.390	0.578	**0.950**

Complexity: $0.380 \rightarrow 0.857$. Verbosity: $0.390 \rightarrow 0.950$. The two worst heads on `hh-rlhf` become near-perfect on targeted pairs — on an independent dataset. The model knew what these concepts meant all along.

Secondary finding — helpfulness-targeted pairs ($N = 200$).

Non-helpfulness heads outperform the helpfulness head on pairs constructed to vary helpfulness. This is Analysis 1's entanglement in practice — a shift in helpfulness activates safety, harmlessness, and instruction_following simultaneously. Weight vectors are orthogonal in W, but they act on a backbone H where these concepts were never disentangled.

Readability collapses on every TYPE B subset — its `hh-rlhf` score was spurious correlation with the holistic

TABLE VI: Head accuracy on helpfulness-targeted TYPE B pairs.

Head	Accuracy
engagement	0.720
safety	0.710
instruction_following	0.710
harmlessness	0.710
helpfulness (matching)	0.670

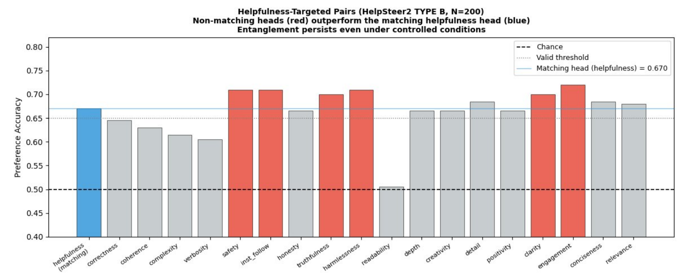

Fig. 6: Head accuracy on helpfulness-targeted TYPE B pairs ($N \approx 200$). Non-matching heads consistently outperform the helpfulness head — backbone entanglement persists under controlled evaluation.

signal annotators preferred in that mixed pool.

VI. What This Means

The failure on `hh-rlhf` is a *measurement failure*, not a model failure. When evaluation data isolates a single dimension, the heads work — strongly, predictably, on a dataset the model never saw. The three datasets measure fundamentally different things:

- **hh-rlhf:** can the head recover holistic human judgment where all dims co-vary? Mostly: no.
- **Reward Bench:** can it recover capability-specific judgments? Mostly: yes.
- **HelpSteer2 TYPE B:** can it recover judgment when exactly one dim varies? Yes, strongly.

Same model. Same weights. Completely different picture.

VII. The Safety Implication

The safety head is real — 0.877 probe accuracy, 0.87–0.95 on refusals subsets. But auditing it requires evaluation data that isolates safety as the varying dimension. On `hh-rlhf` it looks like noise (0.520).

A safety team auditing on naturalistic preference data sees the safety head at chance and draws one of two conclusions: the model encodes unsafe behavior, or auditing is fundamentally impossible. These results suggest a third explanation: the evaluation protocol was never set up to reveal the signal. *How* you test matters as much as *what* you test.

VIII. The Harder Problem

ArmoRM is the easiest case — explicitly labeled bases, by designers, upfront. Reliable validity still required three datasets and controlled per-dimension pair construction. Approaches like LoRe [**?**] learn basis dimensions in a purely data-driven way, without labels. With ArmoRM, you start with a name and build pairs to test it. With a latent basis, the name must first be generated. Directions:

- **User weight inspection:** find users with near one-hot weights over basis dims; read their preference histories.
- **Activation on known stimuli:** vary one concept deliberately; see which dim activates most.
- **Correlation with characterised heads:** use ArmoRM heads as a reference frame for latent activations.

The methodology transfers; the difficulty of generating the initial hypothesis does not.

NOTES

1) The `hh-rlhf` train split pools both harmlessness and helpfulness subsets. The harmlessness head at 0.525 on harmlessness-subset pairs specifically is a stronger failure than the headline number.
2) Helpfulness at 0.429 is noise, not inverse measurement. HelpSteer2 confirms recovery to 0.670 on targeted pairs.
3) Complexity pair count is small ($N = 35$; 95% CI: 0.74–0.97). Verbosity ($N = 200$) is the more reliable anchor.